

AGCS-21701: DATA SCIENCE

CO-1: Introduction to Data Science — Complete Exam Notes

Covers: Evolution, Need & Components of Data Science | Data Science Process | DS vs BI | Applications | Ethical Issues | Privacy & Security | Responsible Data Use

1. What is Data Science? (Definition)

"Data Science is an interdisciplinary field that combines scientific methods, statistical techniques, computational algorithms and systems to extract knowledge, patterns and actionable insight from data — structured or unstructured — in order to support better decision-making."

- Structured data = organised in rows/columns (spreadsheets, relational DB).
- Unstructured data = text, images, audio, video, sensor streams, social media.
- Interdisciplinary — no single field (statistics / CS / business) covers it alone; draws on all three (Section 3).

2. Evolution of Data Science (6 Stages)

1. Data Processing Era (before 1990s) — record-keeping/transaction processing (payroll, inventory); manual analysis; purely operational.
2. Database & Data Warehousing Era (1990s) — RDBMS + SQL; rise of the data warehouse consolidating multi-department data for reporting.
3. Business Intelligence Era (late 1990s–2000s) — dashboards, scorecards, OLAP cubes; KPI-tracking 'data-informed manager'; purely descriptive.
4. Big Data Era (2000s–2010s) — internet/e-commerce/mobile/IoT explosion; characterised by the 3 Vs — Volume, Velocity, Variety (sometimes + Veracity, Value); needed distributed storage & processing.
5. Data Science Era (2010s–present) — statistics + ML + domain expertise on big-data tech; shift from descriptive to predictive/prescriptive; 'Data Scientist' emerges as a distinct role.
6. Agentic AI & Physical AI Era (2020s–, emerging) —
 - Agentic AI: LLM-based systems that autonomously plan, decide, and execute multi-step tasks with limited supervision (given a goal, the agent queries data, analyses, and acts).
 - Physical AI: AI embodied in the physical world — robots, autonomous vehicles, smart industrial equipment — processing real-time sensor data (vision, touch, motion) and acting on the environment.

Exam tip: state stages in order; for each give (a) dominant technology and (b) main purpose of data. Overall shift: record-keeping → reporting → descriptive insight → handling scale → prediction/decision automation → autonomous, agentic & physically-embodied action.

3. Need for Data Science

3.1 Data-side Drivers

- Exponential growth of data (transactions, clicks, GPS, sensors) — needs automated/algorithmic analysis.
- Growing variety of data — text, images, audio/video, network/graph data beyond simple SQL.
- Real-time / high-velocity requirements — fraud detection, ad bidding, recommendations need millisecond response.

3.2 Business / Organisational Drivers

- Data-driven decision making instead of intuition.
- Competitive advantage — spotting shifts before rivals.
- Personalisation at scale for millions of customers.
- Prediction & forecasting — churn, failure, demand — enabling pre-emptive action.
- Efficiency & automation of decisions (loan approval, delivery routing).

Worked example: Telecom BI dashboard shows past churn (descriptive). Data science combines call/complaint/payment records to predict which active customers will churn next month → proactive retention offer (pre-emptive action).

4. Components of Data Science

Classic depiction: a three-circle Venn diagram with Data Science at the intersection of all three components.

Components of Data Science — Three-Circle Venn Diagram

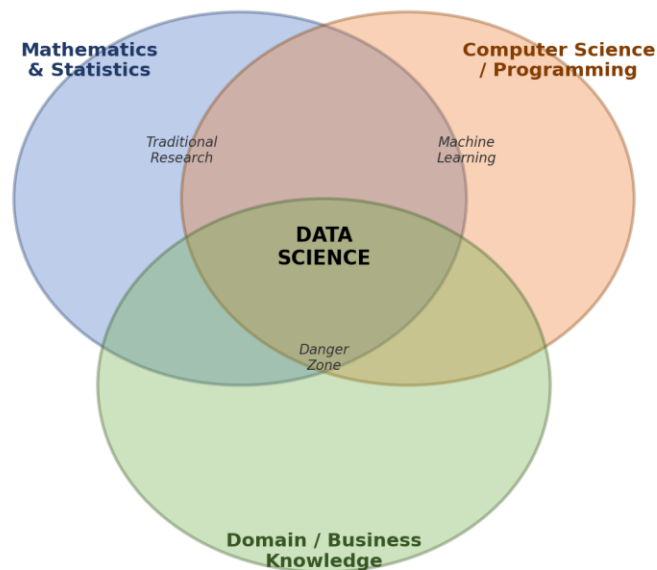


Fig 1: Three-Circle Venn Diagram — Components of Data Science

4.1 Mathematics & Statistics

- Descriptive statistics — mean, median, mode, variance, standard deviation.
- Probability theory — reasoning about uncertainty.
- Inferential statistics — hypothesis testing, confidence intervals.
- Linear algebra & calculus — vectors/matrices, optimising model parameters.

4.2 Computer Science / Programming

- Programming (Python, R, SQL) to collect, clean, transform, analyse data.
- Data structures & algorithms; databases & big-data systems.
- Software engineering practices for production deployment.
- Machine learning implementation.

4.3 Domain / Business Knowledge

- Knowing which questions are worth asking and which data is relevant.
- Interpreting results meaningfully for the specific field.
- Recognising when a statistically valid result is impractical/misleading.

Venn overlaps: Maths+CS (no domain) = Machine Learning | Stats+Domain (no programming) = Traditional Research | Programming+Domain (no stats) = Danger Zone | All three = Data Science.

4.4 Supporting Sub-components

- Data engineering — building pipelines to collect/move/store data.
- Data visualisation — charts, graphs, dashboards.
- Machine learning — models learning patterns from historical data.
- Communication skills — translating technical results for non-technical stakeholders.

5. The Data Science Process (Life Cycle)

A structured, repeatable, iterative sequence turning a business question and raw data into validated, actionable insight.

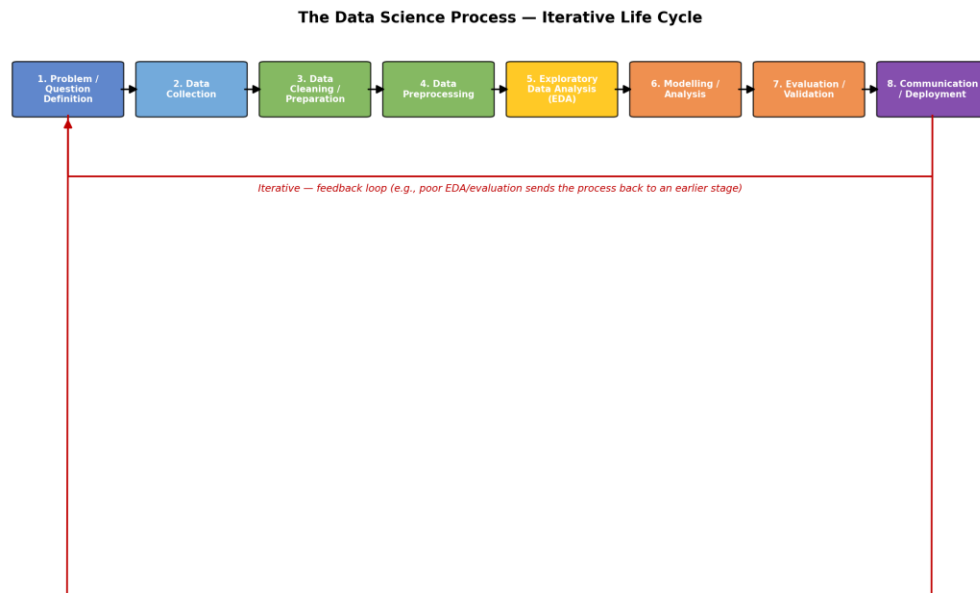


Fig 2: The Data Science Process — Iterative Flow Diagram

7. Problem/Question Definition — precisely define the business question with stakeholders.
8. Data Collection — interviews, questionnaires, databases, web scraping, APIs.
9. Data Cleaning/Preparation — fix errors, inconsistencies, missing values.
10. Data Preprocessing — integration, reduction, transformation, discretisation.
11. Exploratory Data Analysis (EDA) — plots/statistics to study structure, distribution, anomalies.
12. Modelling/Analysis — apply statistical/ML techniques to answer the question.
13. Evaluation/Validation — assess accuracy and generalisation to new (unseen) data.
14. Communication/Deployment — present via reports/dashboards, or deploy into a live system.

Iterative, not linear: EDA revealing data-quality issues loops back to Stage 3 (cleaning); poor evaluation results loop back to Stage 6 (modelling) or even Stage 2 (data collection).

5.1 Why a Defined Process Matters

- Ensures consistency & repeatability across teams/projects.
- Reduces risk of skipping essential steps (cleaning, validation).
- Provides common language for technical & non-technical stakeholders.
- Supports planning — collection + cleaning alone typically take 60–80% of project time.

5.2 Data Science — Features / Advantages / Disadvantages

- Features: interdisciplinary; handles all data types; iterative process; predictive + prescriptive; ML-reliant; produces reusable data products.

- Advantages: enables prediction/forecasting; uncovers hidden patterns; personalisation at scale; improves decision accuracy; adaptable across domains; automates repetitive tasks.
- Disadvantages: high resource/skill needs; time-consuming pipeline; data-quality dependency ('garbage in, garbage out'); black-box interpretability; ethical/privacy risk; infrastructure cost.

5.3 Business Intelligence — Features / Advantages / Disadvantages

- Features: structured internal data (ERP/CRM) in a data warehouse; reporting/dashboard/OLAP focus; short extract-aggregate-visualise pipeline; user-friendly, little coding.
- Advantages: fast visibility into past/current performance; easy to learn/use; faster & cheaper to implement; suits standard recurring reporting.
- Disadvantages: backward-looking only; mostly structured data only; no automatic deep-pattern discovery; depends on well-structured warehouse; limited automation.

6. Difference between Data Science and Business Intelligence

6.1 Three Levels of Analytics

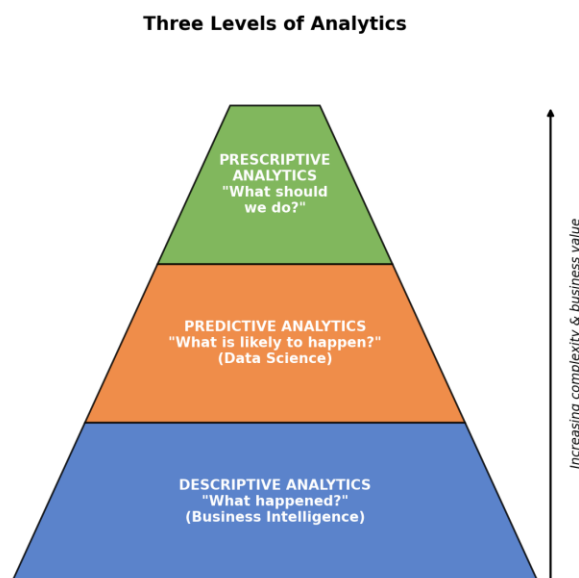


Fig 3: Three Levels of Analytics — Descriptive, Predictive, Prescriptive

- 15.Descriptive analytics — 'What happened?' → core of BI.
- 16.Predictive analytics — 'What is likely to happen?' → core of Data Science.
- 17.Prescriptive analytics — 'What should we do?' → most advanced level of Data Science.

6.2 Detailed Comparison Table

Aspect	Business Intelligence (BI)	Data Science
Primary focus	Descriptive — what happened/is happening	Descriptive, predictive & prescriptive
Type of data	Mostly structured, internal data	Structured, semi-structured & unstructured
Techniques	Reporting, dashboards, OLAP, SQL	Statistics, ML, data mining, advanced analytics
Typical output	Static/interactive reports & dashboards	Predictive models, algorithms, data products
Time orientation	Backward-looking	Forward-looking + historical analysis

Aspect	Business Intelligence (BI)	Data Science
Skill set	Business analysts; SQL/dashboard tools	Statisticians, programmers, domain experts
Process complexity	Short: extract, aggregate, visualise	Full life cycle: collection to deployment
Cost & implementation	Faster, cheaper, minimal coding	Higher cost, longer timelines, specialised skills
Automation level	Human interprets dashboards manually	Can automate the decision itself
Example question	"What were last quarter's sales by region?"	"Which customers will churn, and what offer to send?"

Exam line: BI tells you what happened; Data Science tells you what happened, why, what is likely to happen next, and what to do about it.

7. Applications of Data Science in Various Fields

7.1 Healthcare

Disease prediction & early diagnosis, medical image analysis (X-ray/MRI), personalised treatment plans, readmission-risk prediction, staff/bed-allocation optimisation.

7.2 Finance and Banking

Real-time fraud detection, credit risk scoring, algorithmic trading, customer segmentation, anti-money-laundering pattern detection.

7.3 E-commerce and Retail

Recommendation systems, dynamic/personalised pricing, demand forecasting, review sentiment analysis, market-basket analysis.

7.4 Transportation and Logistics

Route optimisation, predictive vehicle maintenance, ride-hailing demand/surge prediction, autonomous-vehicle perception systems.

7.5 Manufacturing

Predictive maintenance of machinery, automated visual quality control, supply-chain optimisation.

7.6 Government and Public Policy

Census/demographic analysis, crime-pattern prediction, urban/traffic planning, disaster resource distribution, public-health surveillance.

7.7 Education

Learning analytics, personalised/adaptive learning paths, performance prediction, at-risk-student identification.

7.8 Social Media and Entertainment

Content recommendation, sentiment/trend analysis, targeted advertising, audience engagement analysis.

Across all fields, the same data science process (Section 5) applies — only the data, domain knowledge and specific question change.

8. Ethical Issues in Data Science

- Bias & fairness — models reflect real-world biases present in training data (e.g., under-represented groups in historical hiring data).
- Informed consent — individuals should know when/why data is collected and be able to agree or refuse.
- Transparency & explainability — 'black-box' models make decisions hard to understand or challenge.
- Accountability — clarity on who is responsible when a data-driven decision causes harm.
- Data ownership — questions over who owns collected data and what rights each party has.
- Misuse of insights — patterns about habits/vulnerabilities could be exploited if boundaries aren't respected.

Illustrative scenario: A ride-hailing app's demand model, trained only on historical data from a disadvantaged neighbourhood with low app usage, may under-allocate drivers there in future — reinforcing inequality despite being technically 'correct'.

9. Data Privacy and Security

9.1 Data Privacy

Appropriate collection, use, sharing and disclosure of personal/sensitive information, giving individuals meaningful control.

- Data minimisation — collect only what is necessary for a stated purpose.
- Anonymisation / pseudonymisation — mask identifying details.
- Transparency — clear privacy notices.
- Individual rights — access, correction, deletion of one's data.

9.2 Data Security

Protecting data from unauthorised access, alteration, loss or destruction — a technical safeguard underpinning privacy.

- Access control — principle of least privilege.
- Encryption — data at rest and in transit.
- Authentication & authorisation.
- Monitoring and auditing.
- Backup and disaster recovery.

9.3 Privacy vs. Security — Comparison

Aspect	Data Privacy	Data Security
Primary concern	Appropriate collection, use, sharing of data	Protection of data from unauthorised access/loss
Focus	Rights of the individual described by data	Technical safeguards protecting the data itself
Typical measures	Consent, transparency, anonymisation, minimisation	Encryption, access control, authentication, monitoring
Governing question	'Should this data be used this way, with agreement?'	'Can an unauthorised party access this data?'

A dataset can be secure but not private (well protected yet misused), and vice versa — both dimensions are necessary together, not interchangeable.

10. Responsible Data Use

The practical bridge between ethical principles (Section 8) and privacy/security safeguards (Section 9), applied at every stage of the data science process.

- Purpose limitation — use data only for the originally disclosed purpose.
- Fairness — regularly check and correct bias in data/models.
- Transparency — be open about how data is collected, processed and used.
- Accountability — own the outcomes of data-driven systems.
- Minimisation & proportionality — collect/retain only what is needed, only as long as needed.
- Respect for data subjects — treat people as individuals with rights, not anonymous rows of numbers.

11. Glossary of Key Terms

Term	Definition
Data Science	Interdisciplinary field using scientific methods, statistics, algorithms and systems to extract insight from data.
Data Warehouse	Central repository consolidating data from multiple systems, organised for querying and reporting.
OLAP	On-Line Analytical Processing — technology for multi-dimensional analysis (slicing/dicing dashboards).
Big Data	Data characterised by high volume, velocity and variety, exceeding traditional processing capability.
Descriptive Analytics	Analytics answering "what happened" — core of BI.
Predictive Analytics	Analytics answering "what is likely to happen" — core capability of data science.
Prescriptive Analytics	Analytics answering "what should we do" — most advanced level of data science.
Data Privacy	Appropriate collection, use, sharing and disclosure of personal data, respecting individual rights.
Data Security	Protection of data from unauthorised access, alteration, loss or destruction.
Bias (in data/models)	Systematic unfairness in data or model outcomes, often reflecting real-world inequalities.
Agentic AI	AI systems that autonomously plan, decide and execute multi-step tasks with limited supervision.
Physical AI	AI embodied in and interacting with the physical world (robots, autonomous vehicles, smart equipment).

12. Quick Revision Summary

- Evolution: data processing → data warehousing → BI → big data (3 Vs) → data science → agentic/physical AI (emerging).
- Need: exponential data growth, variety, real-time demands, data-driven decisions, competitive advantage, prediction, automation.
- Components: statistics/mathematics + CS/programming + domain knowledge (Venn diagram, Fig 1).
- Process: problem definition → collection → cleaning → preprocessing → EDA → modelling → evaluation → communication (iterative, Fig 2).
- DS vs BI: three levels of analytics — descriptive (BI) → predictive → prescriptive (data science) (Fig 3).

- Applications: healthcare, finance, e-commerce, transportation, manufacturing, government, education, social media.
- Ethics: bias/fairness, consent, transparency, accountability, ownership, misuse.
- Privacy vs security: privacy = appropriate use/sharing; security = protection from unauthorised access.
- Responsible use: purpose limitation, fairness, transparency, accountability, minimisation, respect for data subjects.